



Reflect with us - Week 5



This assignment will not be graded and is only for practice.

1 point

Consider a map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, $v = [xy]$,

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\det(A) \neq 0$. Which of the following options are correct?

☐

T is a linear transformation.

☐

T is both one-one and onto.

☐

T is neither one-one nor onto.

☐

T is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is a linear transformation.

T is both one-one and onto.

- Step 1:Step 1:

1 point

What is $T(v_1 + v_2)$?

☐

$A(v_1 + v_2)$.

☐

$Av_1 + Av_2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A(v_1 + v_2)$.

Recall: $A(v_1 + v_2) = Av_1 + Av_2$

- Step 2:Step 2:

1 point

Is $T(v_1 + v_2) = T(v_1) + T(v_2)$?

☐

Yes.

☐

No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Check: $T(cv) = cT(v)$

- Hence T is linear transformation. So Option 1 is true.

- Another method: Write the definition of T explicitly as,

$$T(x, y) = (ax + by, cx + dy)$$



- Step 1: Step 1: What is $T((x_1, y_1) + (x_2, y_2))T((x_1, y_1) + (x_2, y_2))$?

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2) &= (a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) \\ &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\ &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) = T(x_1, y_1) + T(x_2, y_2) \end{aligned}$$

$$(a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) = (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) = (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) = T(x_1, y_1) + T(x_2, y_2)$$

- Step 2: Step 2:

1 point

Is $T(c(x, y)) = cT(x, y)T(c(x, y)) = cT(x, y)$?

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

- Hence TT is linear transformation. So Option 1 is true.

Now in the given problem $T(v) = 0T(v) = 0$, implies $Av = 0Av = 0$. This is the matrix representation of the system of linear equations:

$$\begin{aligned} ax + by &= 0 \\ cx + dy &= 0 \end{aligned} \quad \text{or} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- Step 3: Step 3:

1 point

If the system of linear equations $Av = 0Av = 0$ has a unique solution, then what is the solution?

☐

$v = 0v = 0$

☐

v can be any vector in R^2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$v = 0v = 0$

- Step 4: Step 4:

1 point

What is the condition on AA , for which the system of linear equations $Av = 0Av = 0$ has a unique solution? [Recall from Weeks 1 and 2]

☐

$\det(A) = 0$

☐

$\det(A) \neq 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A) \neq 0$

Concluding, we can say that, if $\det(A)$ is non-zero, then $T(v) = 0T(v) = 0$ implies $v = 0$. Hence TT is one to one.

- Let $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \in \mathbb{R}^2$ $w = [w_1 w_2] \in \mathbb{R}^2$. We are going to find whether there exists a vector $v = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ $v = [xy] \in \mathbb{R}^2$, such that $T(v) = Av = w$.

$Av = w$ is the matrix representation of the system of linear equation:

$$\begin{aligned} ax + by &= w_1 \\ cx + dy &= w_2 \end{aligned}$$

- Step 5:Step 5:

1 point

If $\det(A) \neq 0$, then what can we say about the solution of the above system of linear equations?

- ☐ No solution.
☐ Unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Unique solution.

Concluding from this, we can say that, if $\det(A) \neq 0$, then T is onto.

Hence, Options 1 and 2 are the correct options.

[Check Answers](#)

Your score is: 0/7